any given point of time and thus the use and operation of all these gadgets simultaneously becomes extremely cumbersome. Due to this universal remote controls, which can control a number of devices at the same time, are becoming increasingly popular. A universal remote control is a device that can send different carrier signals which are programmed so that they mimic the signals sent out by many electronic gadgets. As a result of this, a single remote can be used to control many devices at a time.

7.2. IrDA

IrDA stands for InfraRed Data Association. It defines the standards related to the transfer of data between two devices using infrared light.

The IrDA defines standards related to the transfer of data between two devices using infrared

Infrared communication is a form of free space optical data communication.

For the devices to communicate with each other, they need to have a direct line of sight. Thus, infrared communication comes under the category of very-short-range communication. There are many kinds of IRDA specifications. These include IrPHY, IrLAP, IrLMP, IrCOMM, Tiny TP, IrOBEX, IrLAN and IrSimple. These specifications are arranged from the lowest layer to the highest layer. This means that for IrLAP to be possible, IrPHY must be present, and so on. Here is a brief description of each of these specifications:

IrPHY (Infrared Physical Layer Specification)

This is the lowest layer of specifications and facilitates speeds between 2.4 kbit/s to 16 Mbit/s. The wavelength of light used lies at 875 +/- 30nm. IrPHY usually operates in a cone with a semi-angle of 15 degrees. Most IrPHY devices ideally operate in the distance range of 5 to 60 cm apart. These need to have a certain minimum irradiance in order to be visible, while at the same time, not being blinding to the other device when